5 Mounting and dismounting *RR 600*

5.1 Inspection of the delivery

The RR600 is already partly pre-assembled when it arrives at site. The top roll with the rolled up curtain, bottom beam and side frames are delivered in a transport rack of wood. If the door has a top roll cover, it is placed in the transport rack of wood. Drive, control unit, motor cover and accessories are housed in a cardboard box on a separate palette.

The door should be checked when it arrives at site respectively on damages which could caused while transport and on completeness by comparison of the parts which are arrived and the list of parts mentioned within the corresponding paperwork.

5.2 Preparation of assemblage

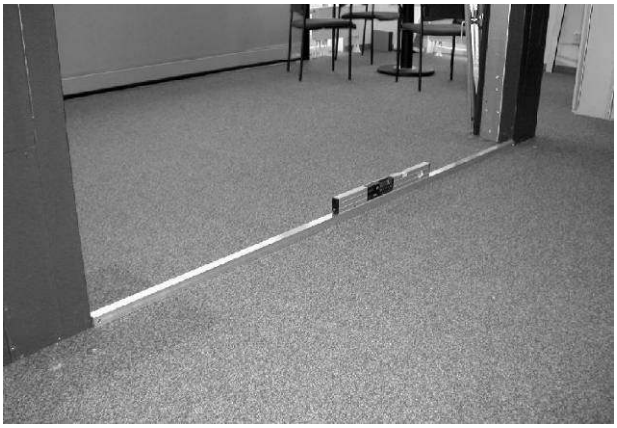
The installation site has to be checked referring the conditions of installation as like as the necessary space, the character of the wall. Perhaps a steel construction or an equivalent construction has to be made in advance. The need is given to fix the columns of the door within the upper area on a solid background like a steel construction or a wall made in concrete.

⚠ **Do not install the doors on isolation panel walls, honeycomb bricks and lightweight partition walls without checking that this is stable enough to support the door, with or without additional reinforcements.**

The light opening dimension (width and height) of the door have to be checked and compared with the values given via the corresponding paperwork.

The site of installation must be closed for any traffic during installation.

Before the installation can start the transport cover foil of the package has to be removed.



1. Put the U profile in the opening, it should have the same length as the door's opening width. Check if the floor is in level. If it is not in level, mark on the wall on the **highest** side.



2. Check the door opening width using the U-profile.
Centre the U-profile in the opening and mark on the floor where the door frame will be placed.

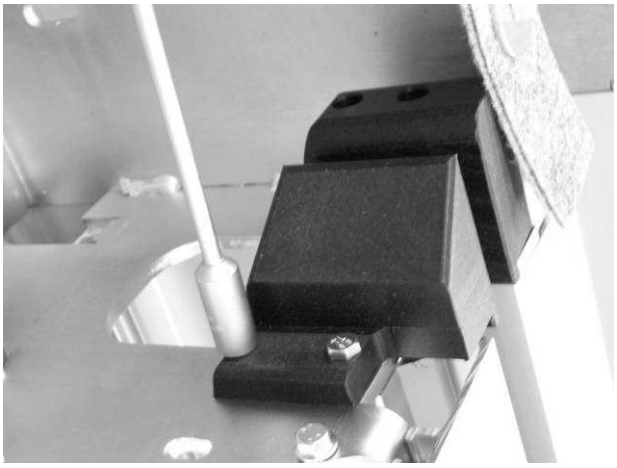


3. Open the covers of the side frames with a screw driver or a rectangle key, which is depending on the design of the door. Remove the installation hatches at the top (directly under the head plates) by loosen the screws a bit. Remember to disconnect the cable of the limit switch on motor side (optional).



Removed installation hatches

4. Slide the screws up and fasten the screws a bit so they don't fall off after you removed the installation hatches.



5. Remove the plastic guides at the front from the head plate.



6. Start with the side frame on the side that is highest. Raise the side frame and place it to fit the mark made on the floor.



7. Fix the side frame temporarily with a screw clamp.



8. Open the low hinge part.



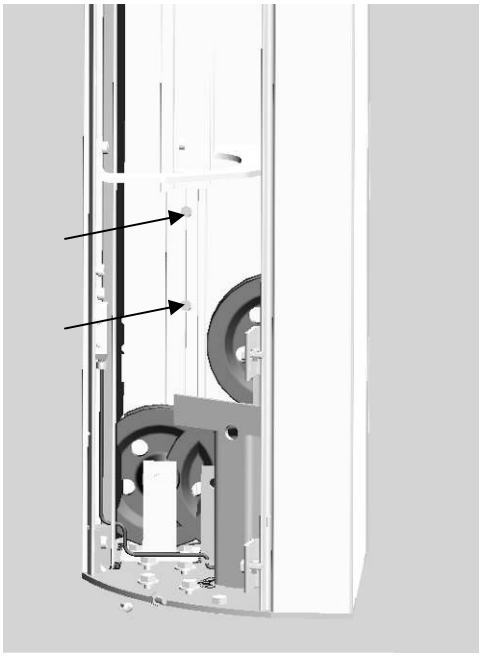
9. Level the door frame vertically in all directions.



10. Drill and fix it with at least three (3) suitable screws at the head plate.



11. Drill and fix it with at least two (2) suitable screws at the bottom.



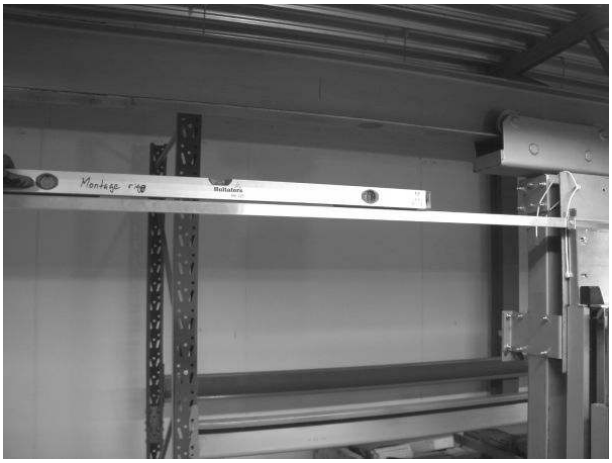
12. If it is not possible to fix to the floor, you can use the pre-drilled holes in the door frame.
Make sure that the wall has a plane surface in order to prevent the frame from bending.



13. Raise the other door frame and fix it temporarily with a screw clamp.



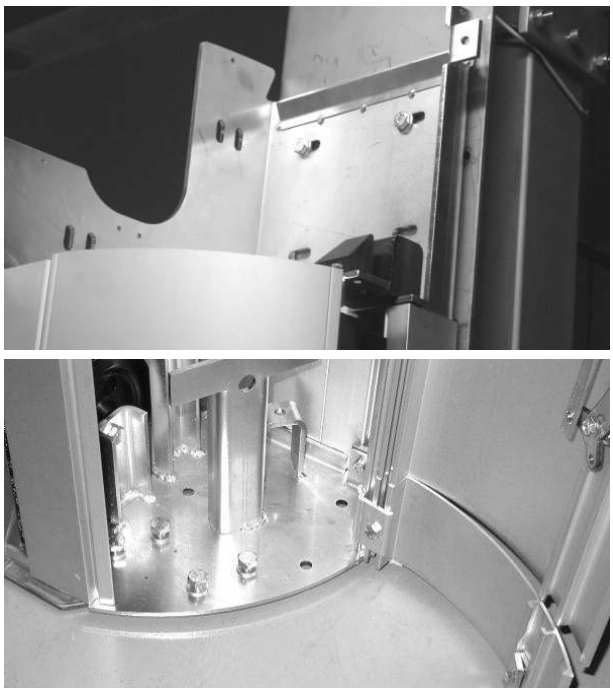
14. Fit the U-profile between the two side frames with 2 screws (M8x16). Check that the distance between the two frames is according to the day light width of the door.



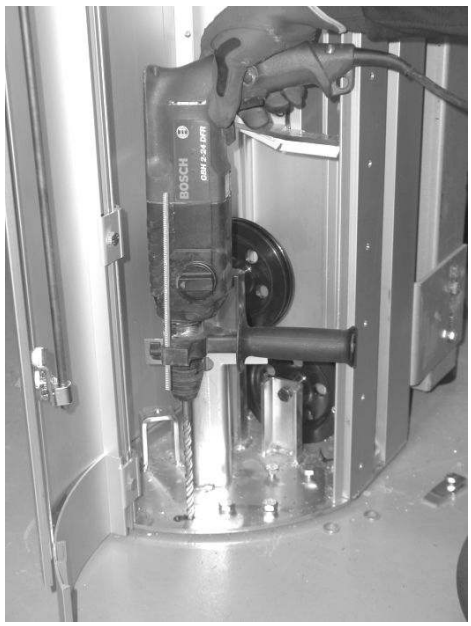
15. Check that the two frames are in level horizontally. You can do that with a spirit level on the U-profile. Use a 1.5 m leveller and place it in the centre. If nessesery leave the required distance under the bottom plate.



16. Level the second door frame vertically in all directions. Be sure it fits to the mark you made on the floor erlier.



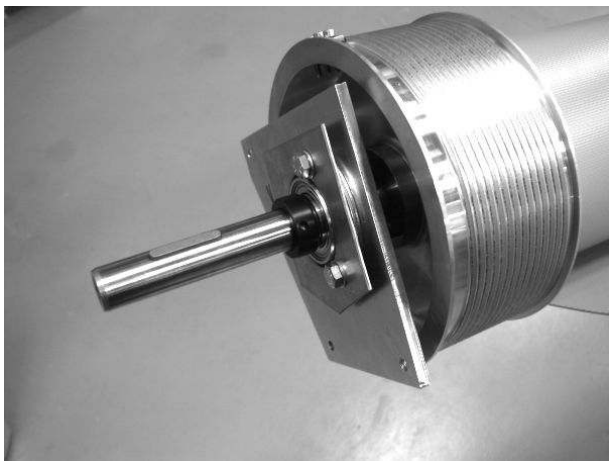
17. Drill and fix it with at least three (3) suitable screws at the head plate (shown in Picture top). Open the low hinge part (shown in Picture bottom).
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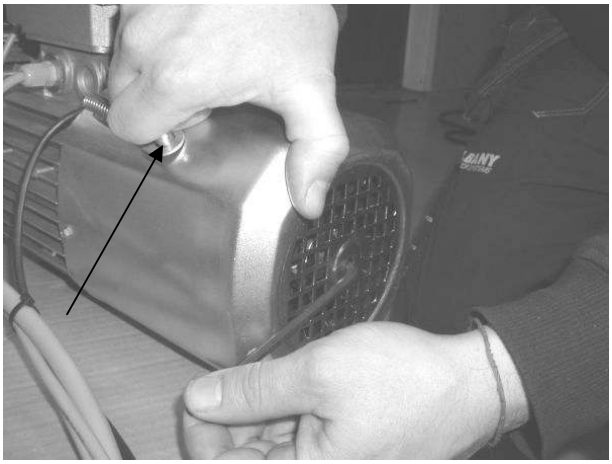
18. Drill and fix it with at least two (2) suitable screws at the bottom.



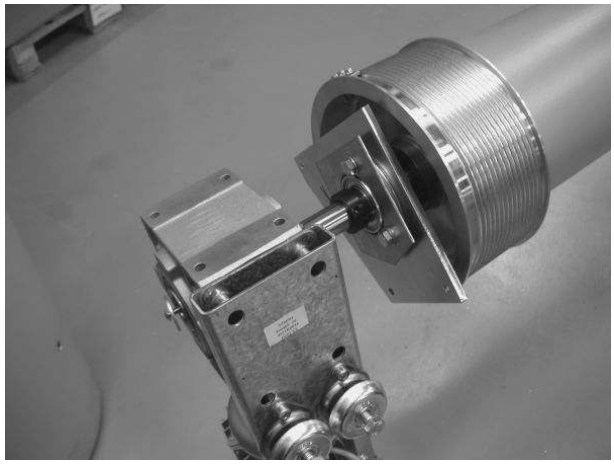
19. Remove the plastic around the top roll. Put the top roll on a pallet, use a suitable lifting device e.g. fork lift. Secure the door blade with a strap. Make sure that the top roll cannot fall down when lifting.



20. Make sure that the pre-installed bearing bracket not is final tighten in order to make it easier to fit in to the top wall bracket. The screws should stay in the brackets.

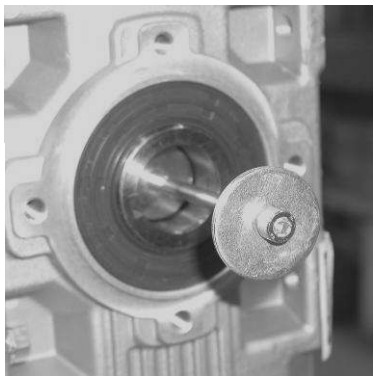


21. Prepare the drive unit by releasing the brake and turn it with an Allen key in order to get the hollow shaft key in right position.

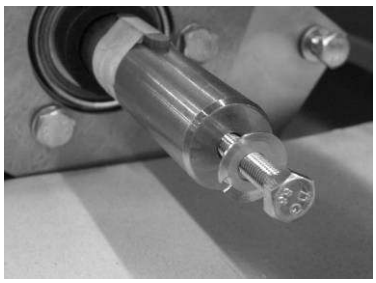


22. Slide on the drive unit onto the shaft.

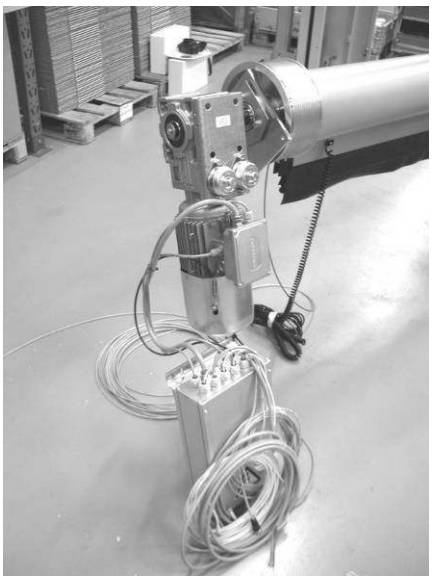
Stub axle with drive unit, not SRD



Stub axle SRD



23. Note! Put on the temporary security screw (M6x70 mm) and washer to prevent the drive unit from falling down during installation.



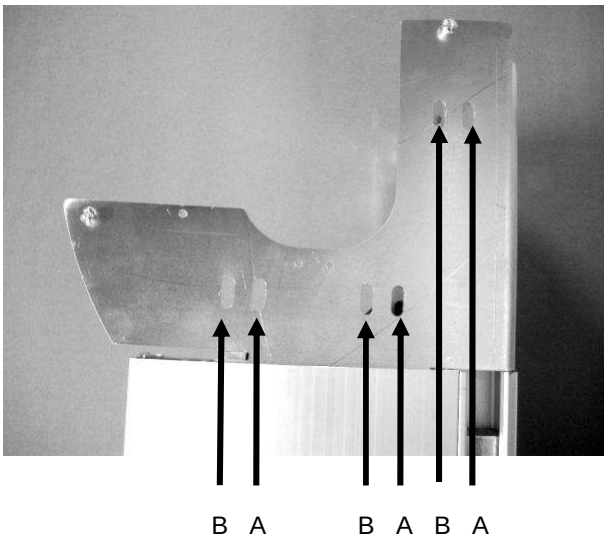
24. The MCC unit is secured to the drive unit with a plastic strap to take the load off the cables during installation; this will be cut away when the MCC unit is screwed to the side frame. Some cables are pre-installed in the side frames.



25. Lift the top roll using a suitable lifting device, e.g. fork lift.



26. Guide the bearing plate onto the wall bracket. Position see Picture 29.



27. Use the inner hole (A) or outer hole (B) depending on which top roll system is used.
A = top roll Ø140 mm, cable drum Ø250 mm
B = top roll Ø194 mm, cable drum Ø197 mm



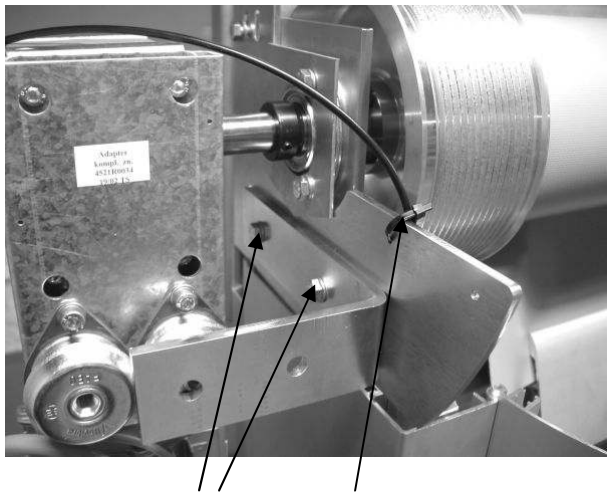
28. Using the motor bracket as a crowbar in a slot of the bearing plate makes it easier to guide the top roll bearing plate in position.



29. Picture is showing the bearing brackets position on the wall bracket.



30. Tighten the screws for the bearing bracket on both sides. 1 x M10x20 and 2 x M10x25 on motor side and 3 x M10x20 on non motor side.



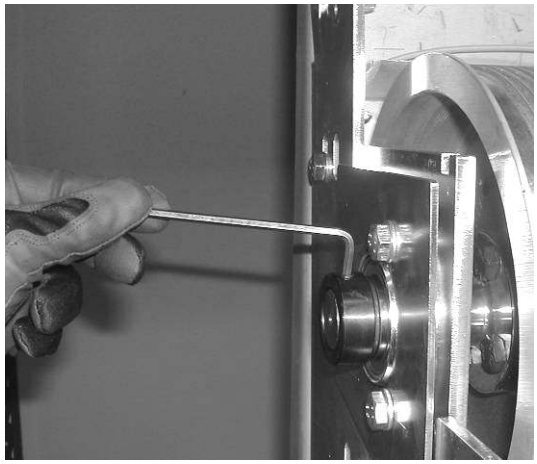
31. 2 x M10x25 at the bottom of the head plate for the motor bracket.
Fix the brake release cable with a cable strap.



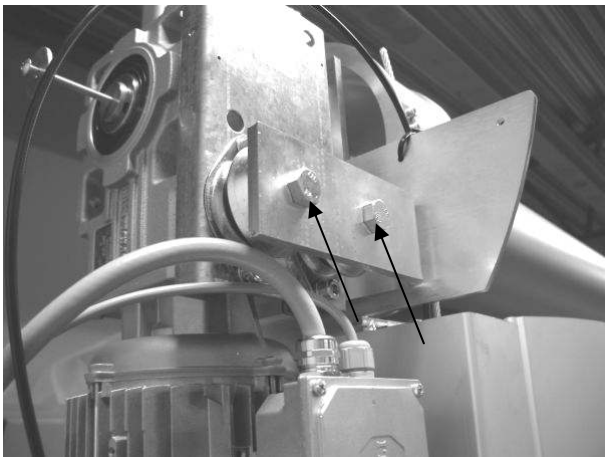
32. Tighten the bearing plate on both sides (M10x25).



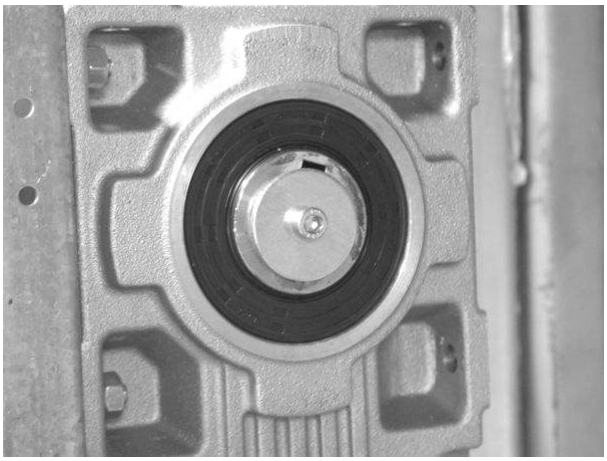
33. Tighten the bearing ring clockwise on both sides. Use a hammer and a drift that fits the hole.



34. Lock the bearing ring on both sides. Use an allen key.

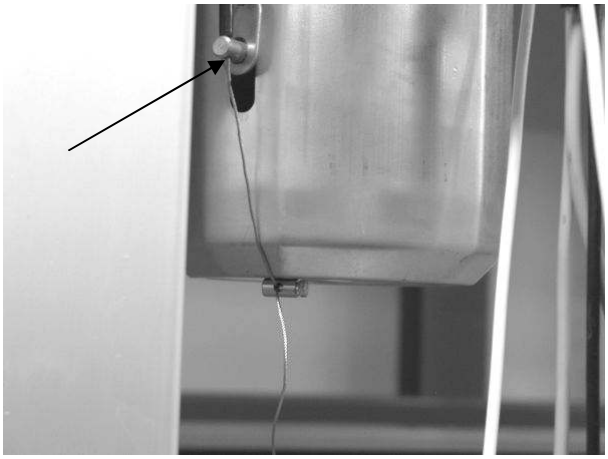


35. Fix the drive unit to the torque arm with two screws (M12x25). Remove the temporary security screw (M6 x 70) and washer (6,4 x 35 x 1,2 mm).

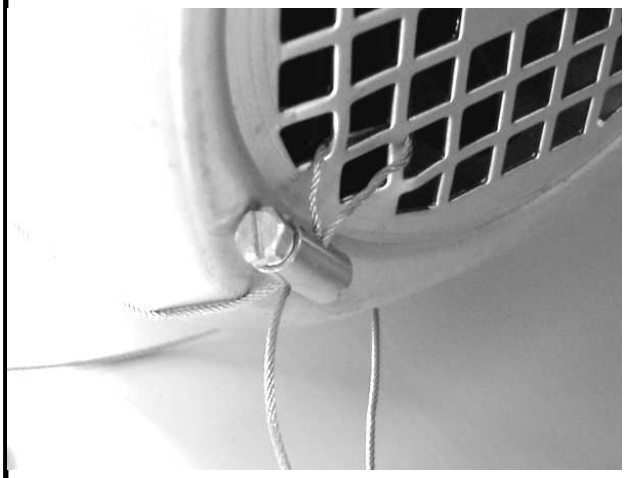


Shows not SRD; if SRD see Picture 85

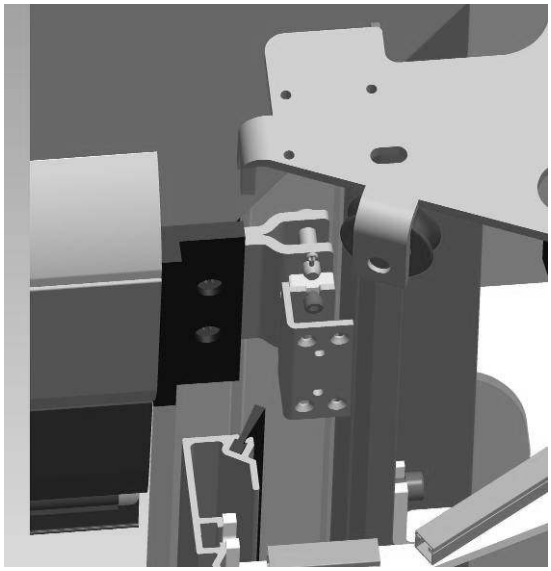
36. Secure the motor to the shaft with screw (M6x16) and washer (6,4 x 35 x 1,2 mm).



37. Install the brake release wire through the brake release shaft.

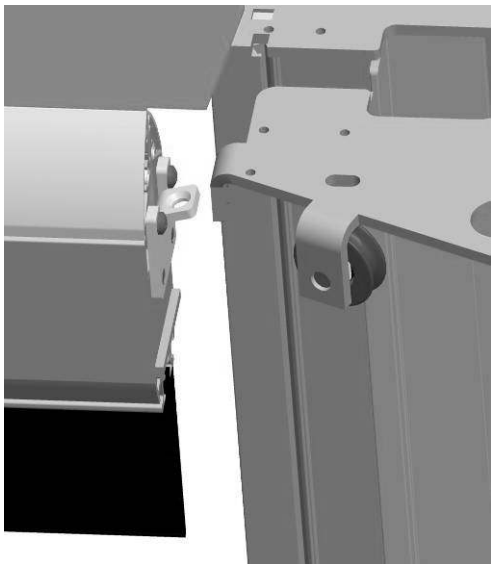


38. Pull the wire around the grating of the motor cover and fix it.
Remove the plastic strap that secured the top roll.



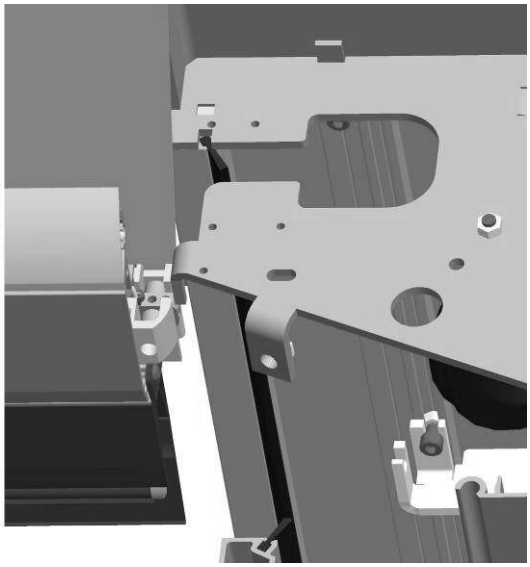
Shows bottom beam standard / pre running photocell at open door

39a. Press in the door blade on both sides into the slot of the top roll bracket. Lift and press to opposite side to make space.
Note: The shape of the door blade may look different depending on door configuration.

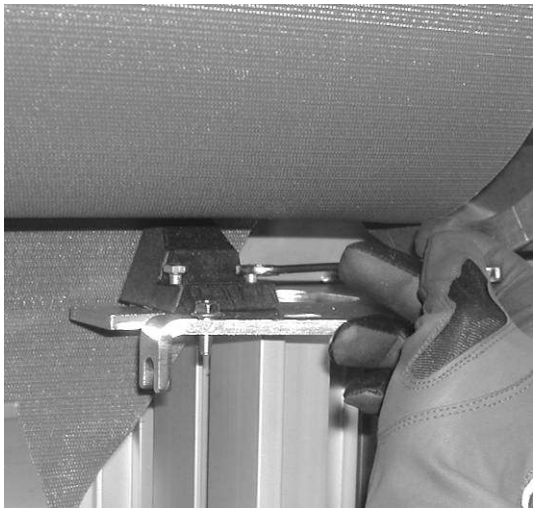


Shows bottom beam crash at open door

39b. Press in the door blade on both sides into the slot of the top roll bracket.

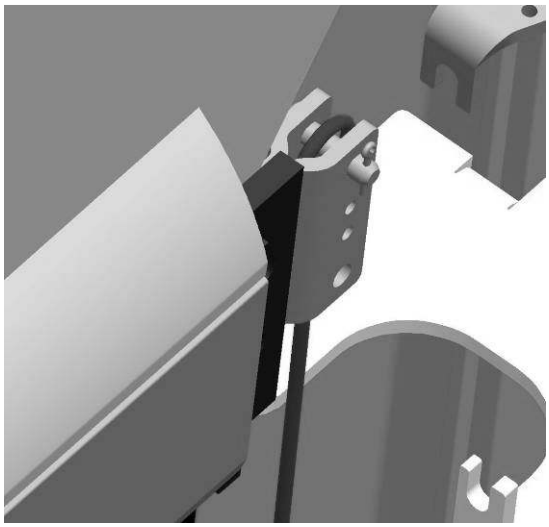


Shows bottom beam SRD at open door

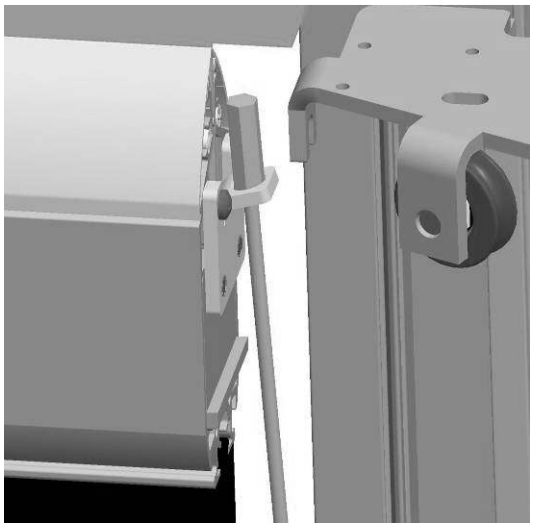


39c. Press in the door blade on both sides into the slot of the top roll bracket.

40. Fit the front plastic guiding block on both sides (left, right) with screws (M8x16).



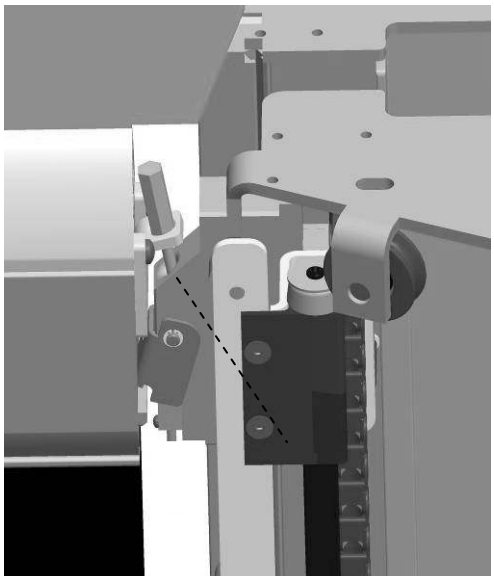
Shows bottom beam standard / pre running photocell at open door
(Shows it without reference plate)



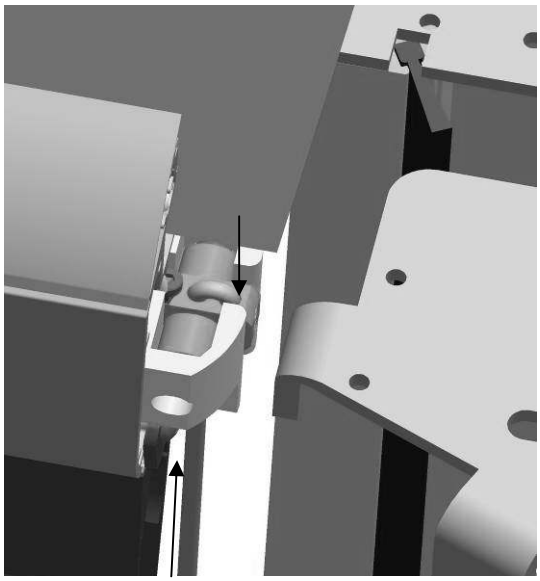
Shows bottom beam crash at open door

41a. Unwind the steel cables and fit them to the bottom beam guide with the steel pin and lock with the safety pin.

41b. Unwind the steel cables and fit them through the holder crash at the bottom beam…



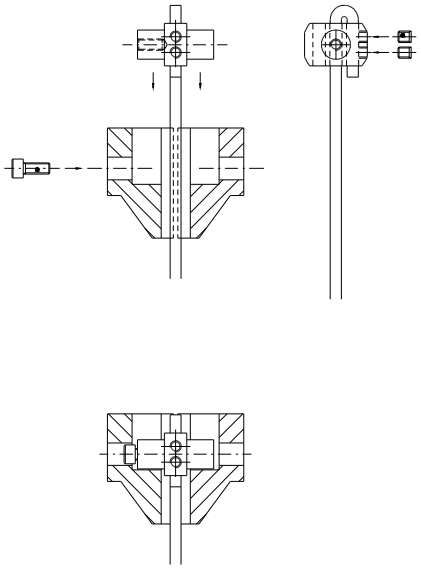
Shows bottom beam crash at open door



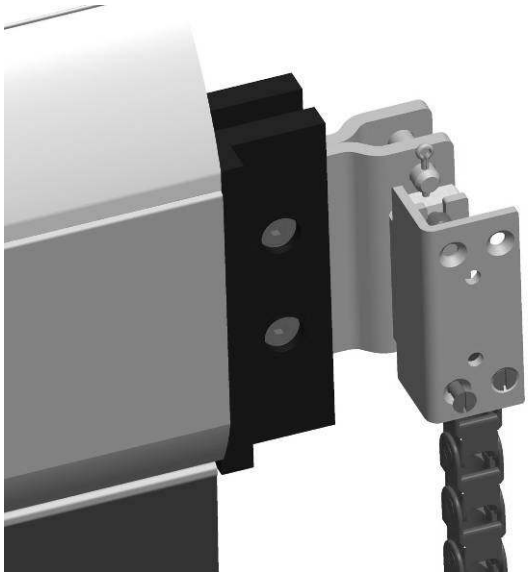
Shows rope lay bottom beam SRD at open door

41c. …the rope must be guided through the wear pad, which is pre-installed in the side frame.
If the door is closed (after installation of control unit), the crash-system will lock in position itself.

41d. Unwind the steel cables and pull it up through the rope guide of the bottom beam (hole in middle). The rope must be pushed down through the hole at the outer side.
Tighten the small screws.



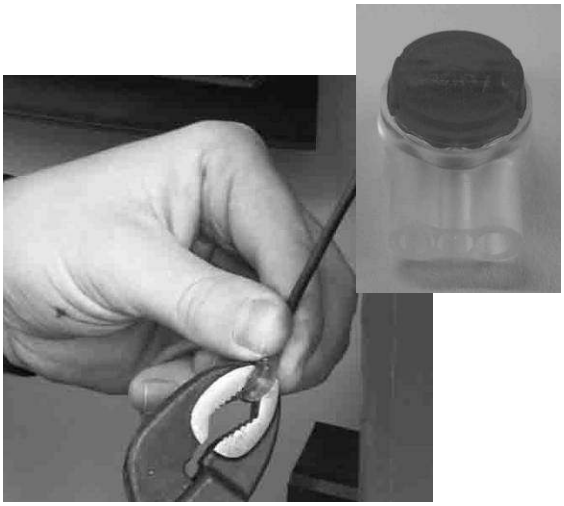
Schematic drawing rope guide at the bottom beam SRD



Shows standard with cable chain

41e.

42. Fasten the cable chain with two screws M3x20 at the reference plate. The reference plate and the cable chain are only on the drive side.



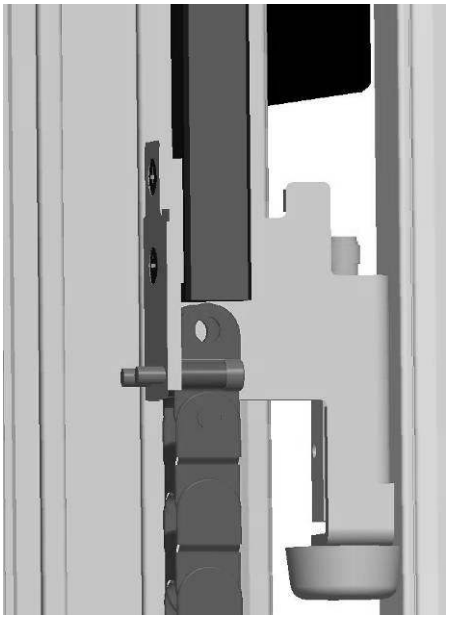
Doors standard



Shows standard with cable chain

43. Install electrical connection between safety edge and cable chain. Do not remove the wire insulation. Press connector together with a pliers.

44. Lay cable as shown



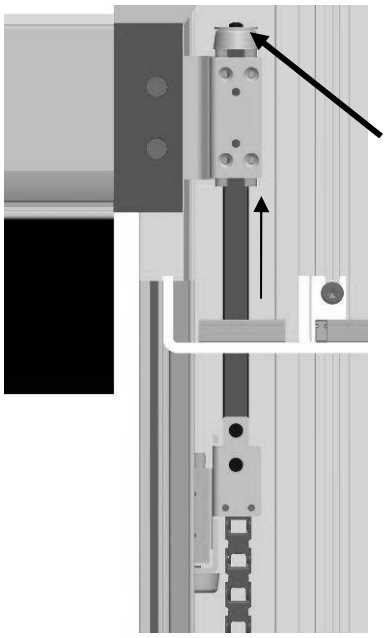
Shows pre running photocell



Shows pre running photocell

45. Screw the cable chain to the bracket of the pre running photocell. Plug the plug on the pre running photocell.

46. Remove the rubber damper from the top of the plastic guide.



Shows pre running photocell

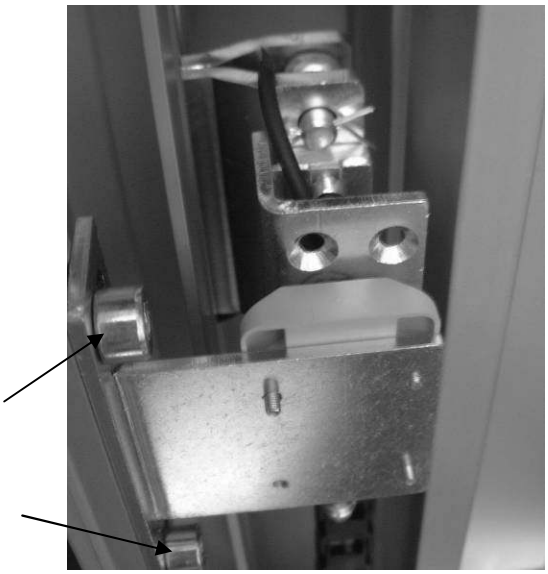


Shows pre running photocell

47. Slide in the plastic guide of the pre running photocell into the guide at the bottom beam upwards. Fix the rubber damper at the top of the plastic guide.

48. Check that the sliding bar is moving easy all the way up and down.
Do the same operation at the other side.





Shows not SRD; if the door is with SRD the upper reference must be adjusted with connected control system: see Picture 95

49. Reinstall the installation hatch (direct under the head plates) on both sides with screws (M8x16). Make sure that the hatch is lining the rest of the door frame profile. Assemble and connect the limit switch if the door has a limit switch in the side frame.

50. The space between the limit switch and the reference plate must be 2 to 5 mm. You can adjust the position of the limit switch by adjusting the bracket. Release the two screws (arrows) to ajust the bracket.



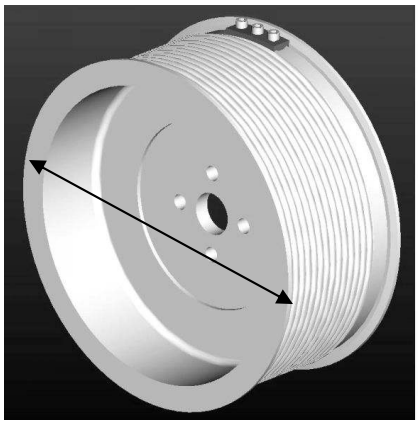
The door can be delivered with different balancing systems depending of the size of the door in combination with different door blades.

To measure the outer diameter of the cable drum makes it easy to detect which system is delivered with your actual door.

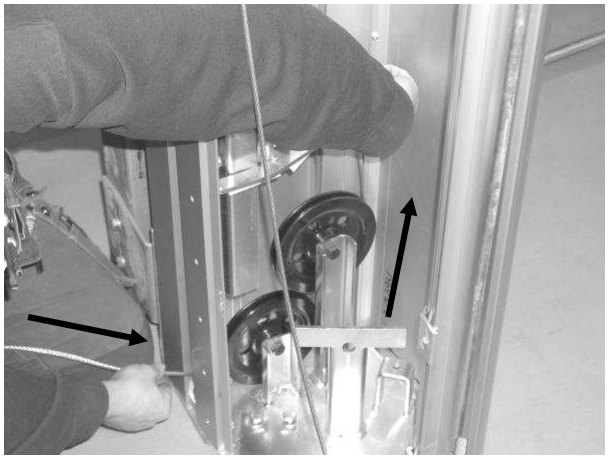
51. Unwind the steel cable on the floor, make sure that there are no kinks or knots.

52.

System A



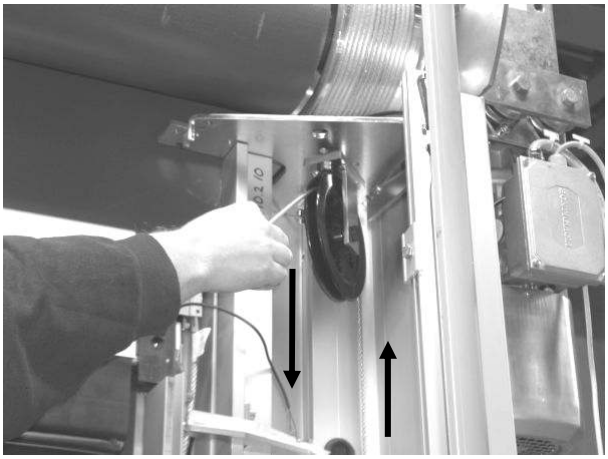
System A = 250 mm



System A

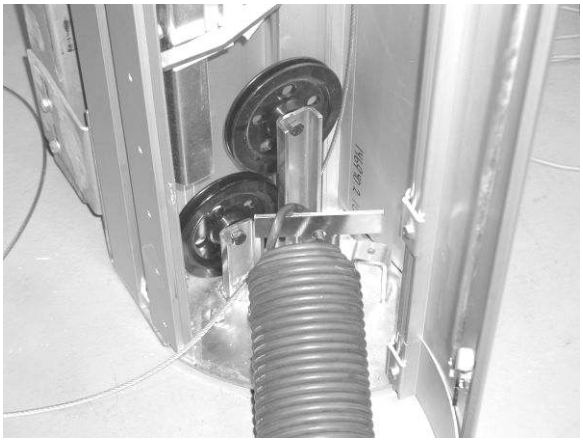
53. Note: The following instruction is for tensioning system A. The cable drum of system A has an outer diameter of 250 mm.
If the door is equiped with tesion system B then go to picture 64.

54. Pull the steel cable that is connected at the bottom profile around the bottom pulleys…



System A

55. … and up around the top pulley…, fit a clamp on the steel cable to keep the steel cable stretched.



System A

56. Shows installed spring. Unpack the spring package. Rise the spring to a vertical position and make sure it not falls out. If the door will have a BBS system, see picture 99.



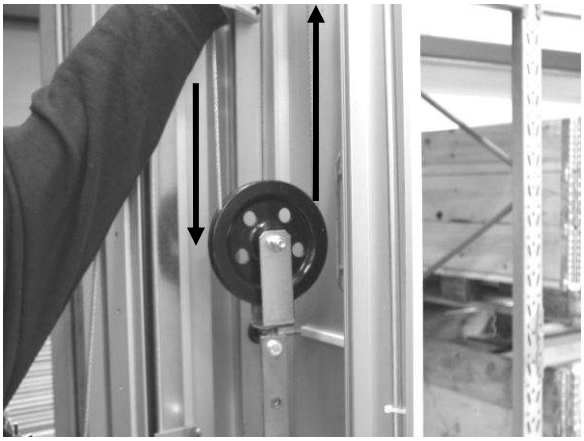
System A

57. Hook up a pulley-block between the top wall bracket and the spring.

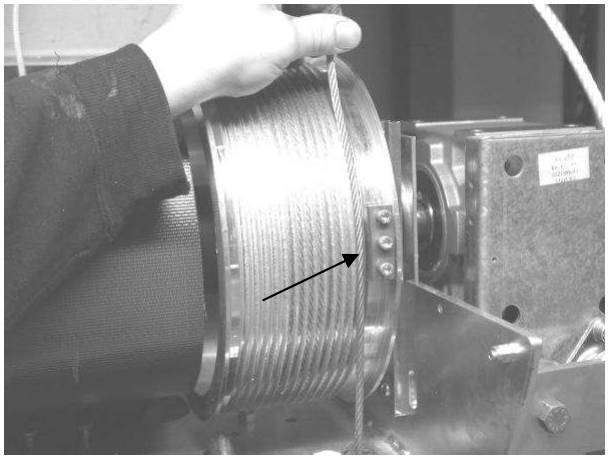


System A

58. Tense the spring with the pulley-block approx. 50mm over the spring label. When the pulley is released and the door has been opened and closed the spring must be positioned like the picture shows. (do not run the door now!)



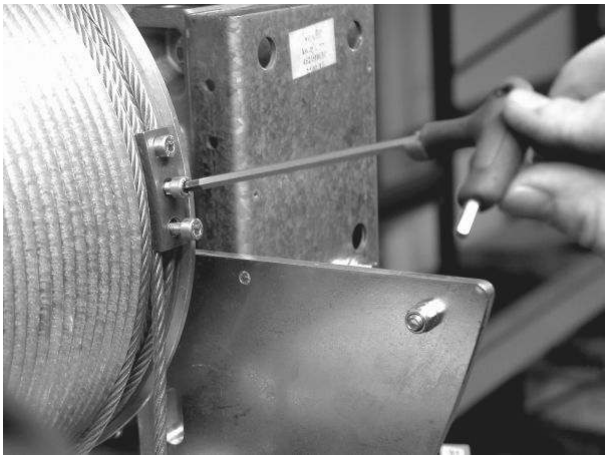
System A



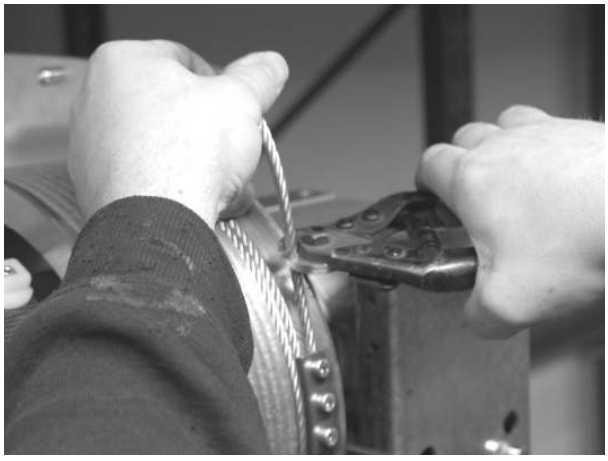
System A

59. Continue with the steel cable from the top pulley down to the spring pulley at the top of the spring. Lay the steel cable around the pulley …

60. … and finally up to the cable drum.
Fit the steel cable on the cable drum directly beside the clamping plate.



System A



System A

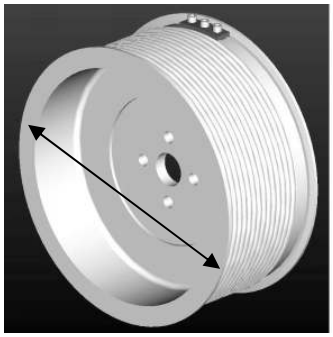
61. Put 3 security turns on the cable drum; pull the steel cable as much as possible. Fix it with the 3 screws. 1 pre-turn have to be seen outside the cable clamp. Note: Make sure that the steel cable is in the slot of the cable drum otherwise it will not wind correctly.

62. Cut the not needed side of the steel cable.



System A

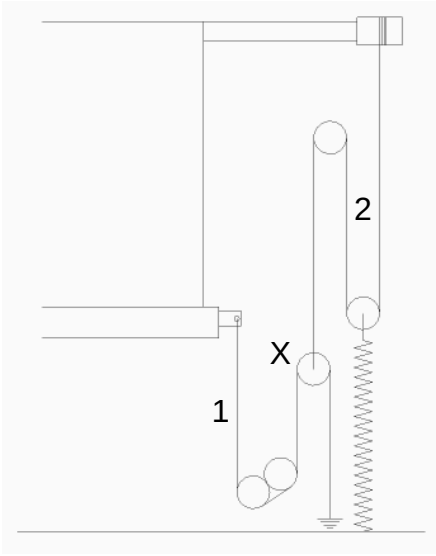
System B



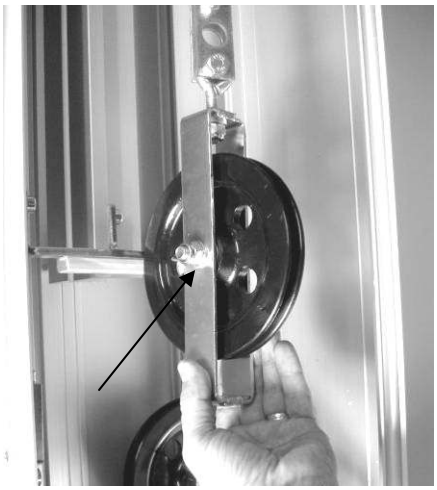
System B = 197 mm

63. Remove the pulley-block. Repeat step 53 to 62 on the other side.

64. Note: The following instruction is for tension system B. The cable drum of system B has a diameter of 197 mm.
If the door is equiped with tension system A then go to picture 53.



Schematic drawing of rope lay
System B



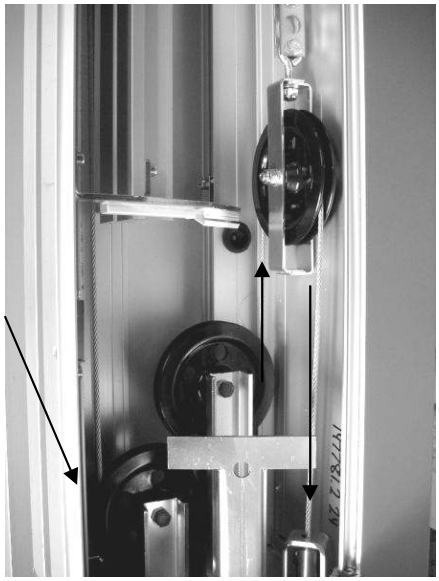
System B

65. The steel rope no.1 is connected to the bottom beam, runs around the pulleys at the bottom plate to pulley X and again to the bottom plate. The steel cable no.2 begins at the pulley X, runs around the pulley at the top of the spring and continues up to the cable drum.

66. Put the reduction pulley X with the attached steel cable no.2 in the side frame. The centre position must be approx. 400 mm from the floor.
Note: The locking nut of the reduction pulley is facing inner wards...



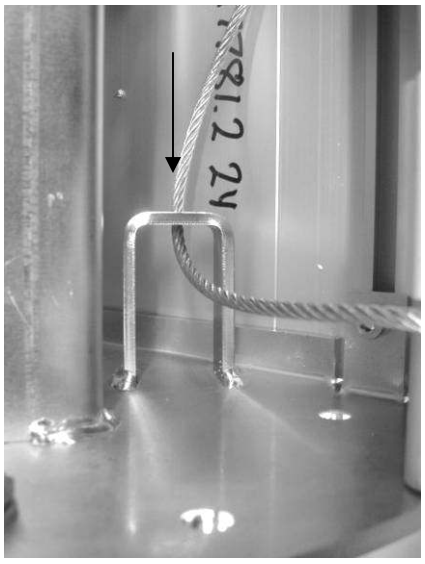
System B



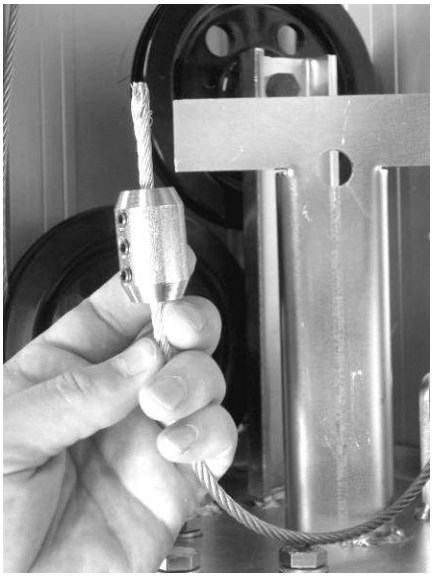
System B

67. …fix the steel cable no.2 temporary against one of the frame reinforcements with a strap or similar.

68. Pull the steel cable no.1 from the bottom beam under the lower pulleys in the frame and up over the loosen reduction pulley X. Then pull it to the holder at the bottom plate…



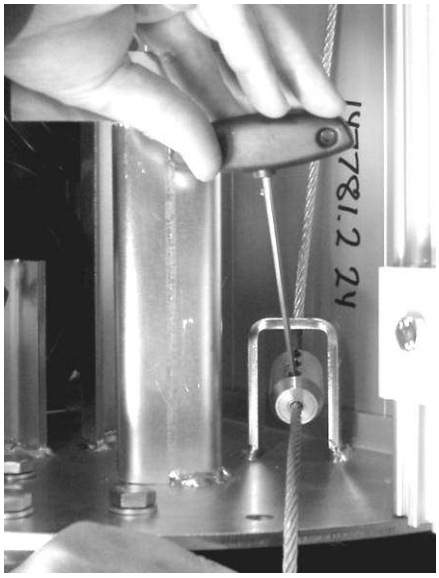
System B



System B

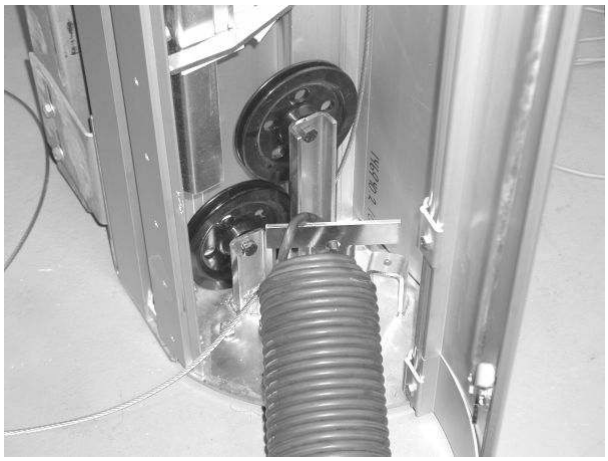
69. … pull the steel cable trough the bottom bracket.

70. Put the cable lock on the steel cable.



System B

71. Stretch the steel cable; make sure that the additional pulley X remains in its position 400 mm from the bottom. Tighten the cable lock with 3 screws and cut the rest of the steel cable.



System B

72. Shows installed spring. Unpack the spring package. If the door will have a BBS system, see picture 99.



System B

73. Rise the spring to a vertical position and make sure it not falls out.



System B

74. Hook up a pulley-block between the top wall bracket and the spring.

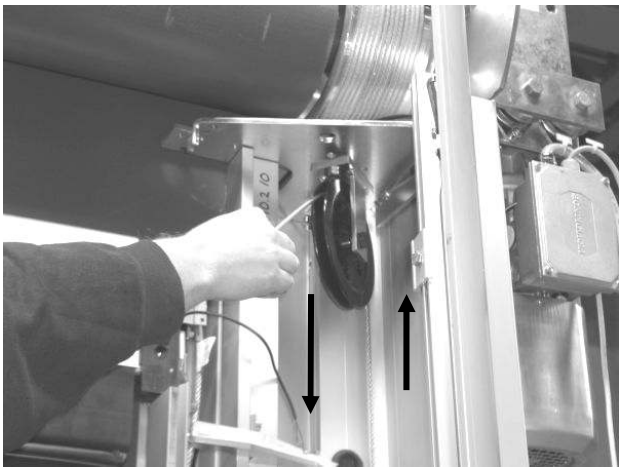


System B

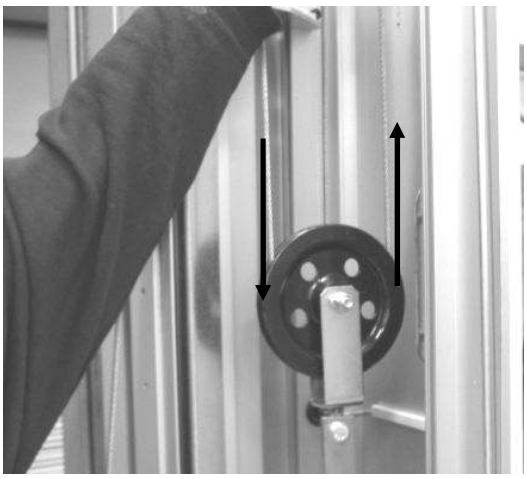


75. Tense the spring with the pulley-block approx. 50mm over the spring label. When the pulley is released and the door has been opened and closed the spring must be positioned like the picture shows. (do not run the door now!)

76. Pull the steel rope no.2 from the loosen pulley X to the head plate…



System B



System B

77. …pull the steel rope around the pulley at the head plate…

78… down to the pulley at the top of the spring.



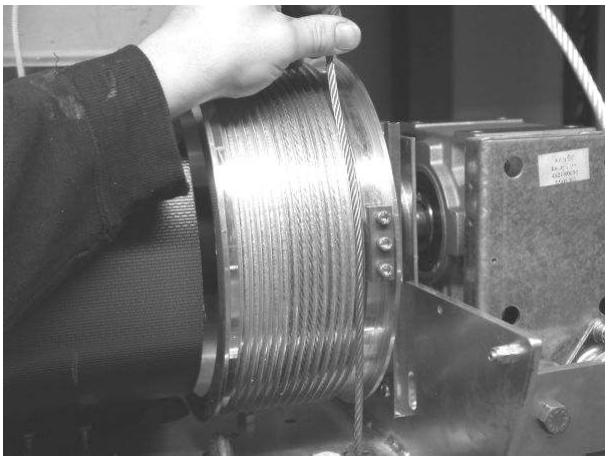
System B



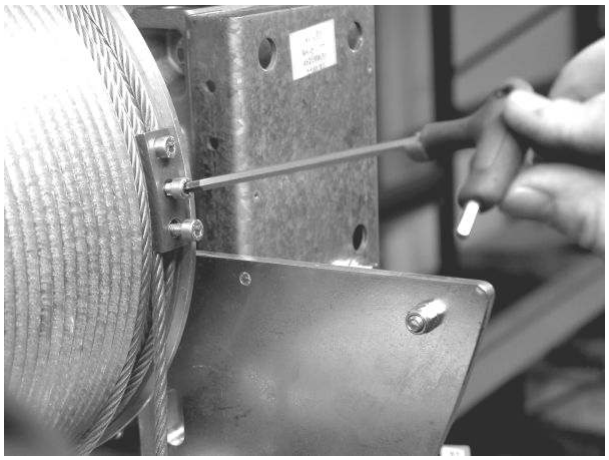
System B

79. Remove the strap at the steel cable no.2 that was fixed temporary. Tighten the steel rope no.2.

80. Check, if the loosen pulley X is at the right position. (Center of pulley approx. 400mm over the floor)



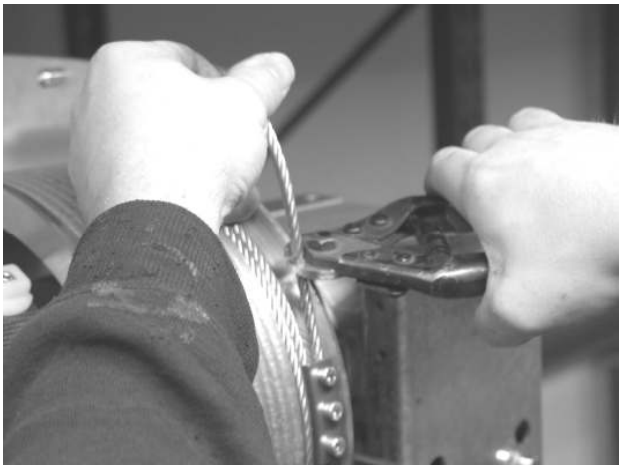
System B



System B

81. Guide the steel cable no.2 up to the cable drum.
Lay the steel cable no.2 on the cable drum directly beside the clamping plate. Pull the steel cable as much as possible.

82. Put 3 security turns on the cable drum. Fix the clamping plate with the 3 screws. 1 turn have to be seen outside the lock.
Note: Make sure that the steel cable is in the slot of the cable drum otherwise it will not wind correctly.



System B

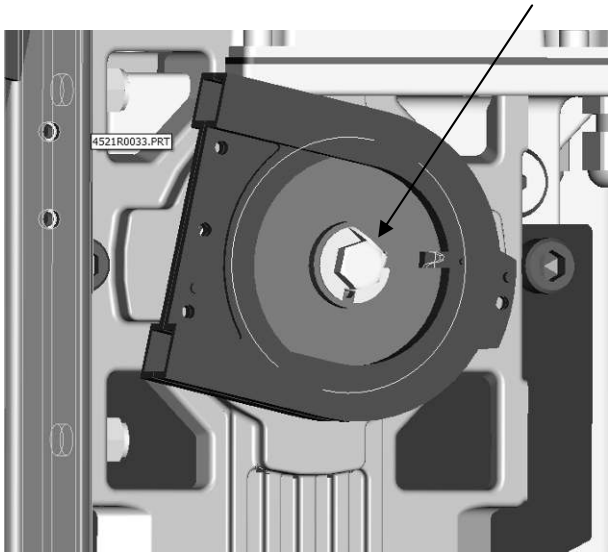
83. Cut the rest of the steel rope.



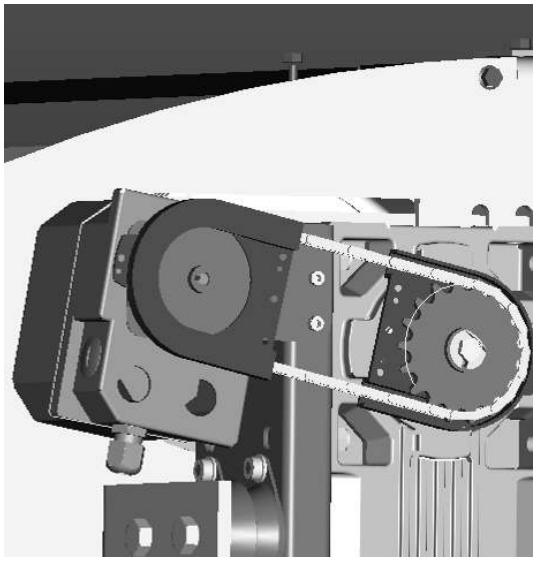
System B

84. Remove the pulley-block. Repeat step 64 to 83 on the other side.

Installation SRD

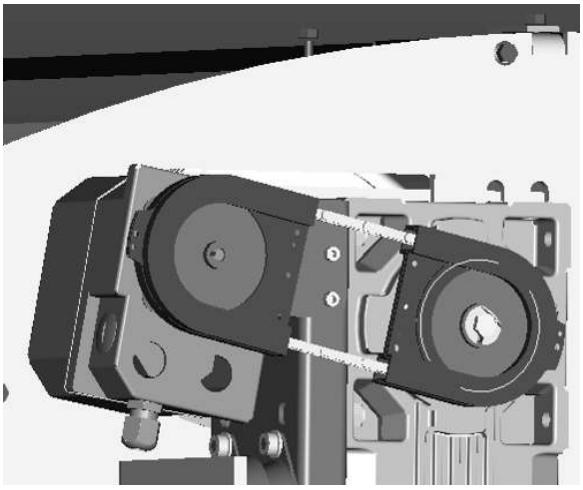


85. Add the chain wheel to the outside of drive unit. Put the screw to the hole in the centre of the shaft and tighten, lock with the lock wascher.



only SRD

86. Remove the plastic chain cover and put the chain over the wheel. Strech the chain and sccrew the limit switch box to the motor bracket with the 2 screws.

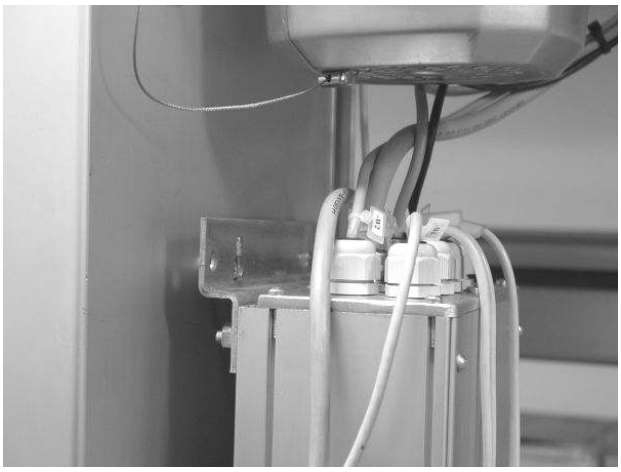


only SRD

Installation of the MCC unit to the side frame

87. Put back the chain cover.
The limit switch inside the box will be set of when the door is in it´s open position and will work as the ref. limit switch.
Go on with Picture 37

88. Remove the security strap between the motor and the MCC unit.



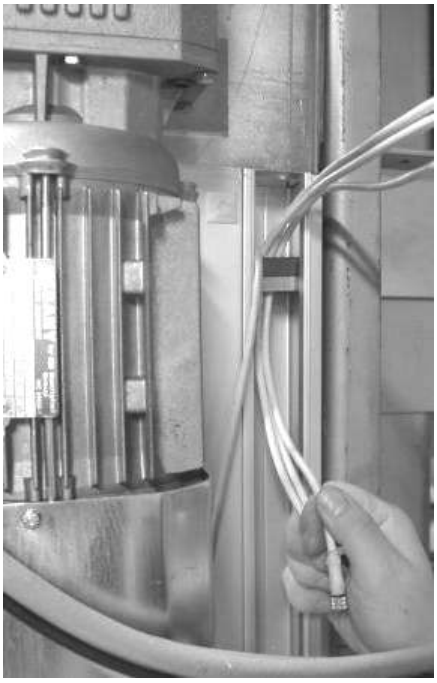


89. Hang up the MCC onto the pre-installed screws (M6x20) on the side of the frame.

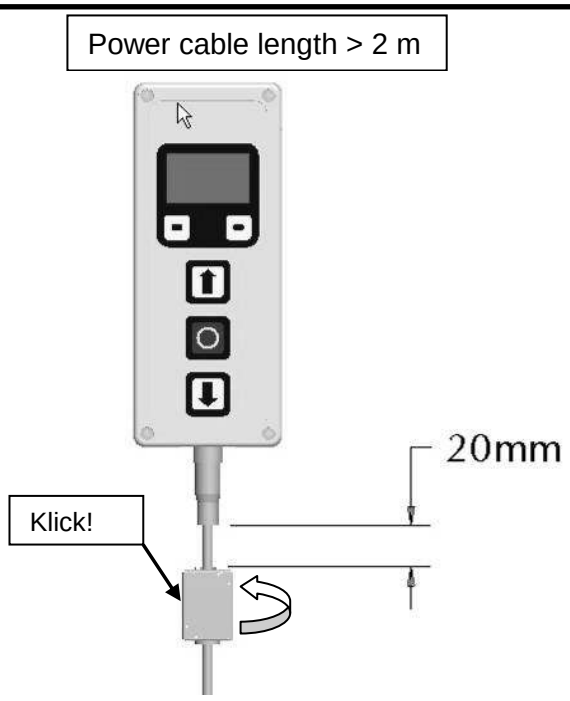
90. Fix the MCC unit to the frame with screws (M6x20).
Remove the plastic cover for the build in cable duct on the frame on both sides.
(arrow).



91. Note:
Each cable is marked in both ends with a tag.



92. Pull all cables trough the cable bracket on the top of the frame and into the cable duct.



92a.

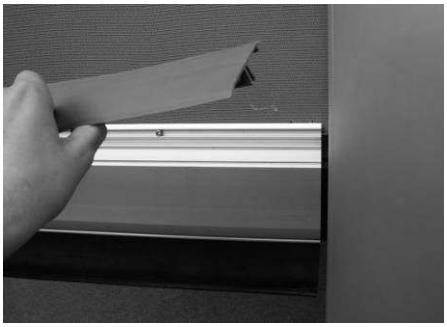
Look in the wiring diagram and make sure that you connect the cables to the correct equipment.
Read the user manual of the MCC.

Set the limits

At doors with a limit switch in the side frame the top position is detected by reference limit switch. (see Picture 50).

Doors with SRD do not have a limit switch in the side frame. The top position is detected by reference limit switch at drive unit (see Picture 95).

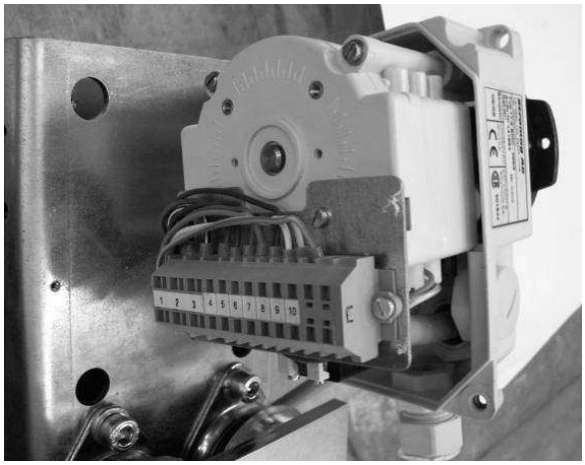
93.





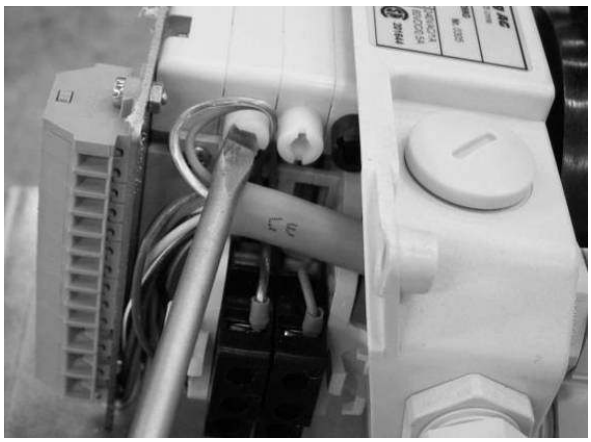
Tighten all screws at bottom profile

94. Pull off the cover profile from the bottom beam. If only the right and the left screw for fastening the bottom profile are assembled, you can adjust the bottom profile to the side after you released the screws. Put in all screws and nuts in every hole of the bottom profile and tighten them. Push in the cover profile.



only SRD

95. Disassemble the cover of the limit switch at the drive unit.



only SRD

96. Adjust the top limit with a screw driver. Check the top limit. Assemble the cover of the limit switch.

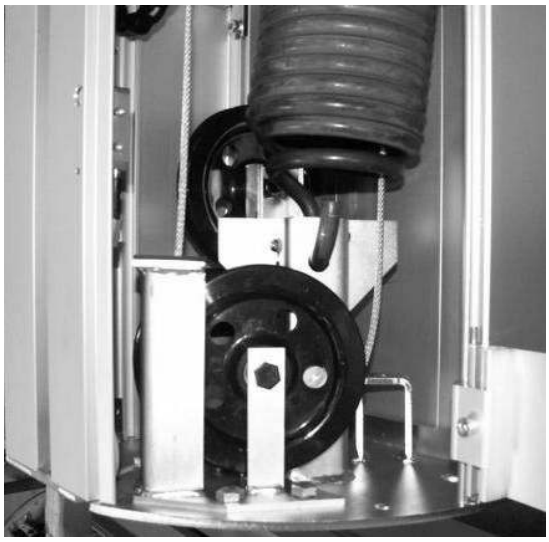


97. Close all cable ducts in the frames with the enclosed covers.

ered with



Installation BBS-System

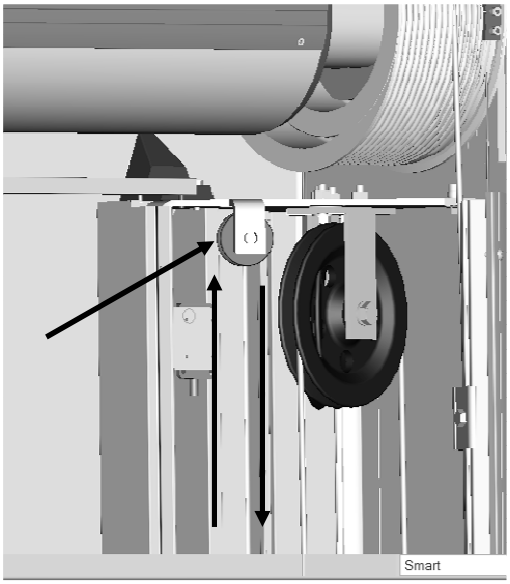


98. If the door has no BBS and no covers, close the covers of the side frames and lock them with a key or screws (depending on version). If the door has a BBS system, see Picture 101. If the door has covers, see Picture 107.

99. The steel rope for BBS is guided around the pulley at the bottom plate already. The steel rope is laid through the spring to the pulley.

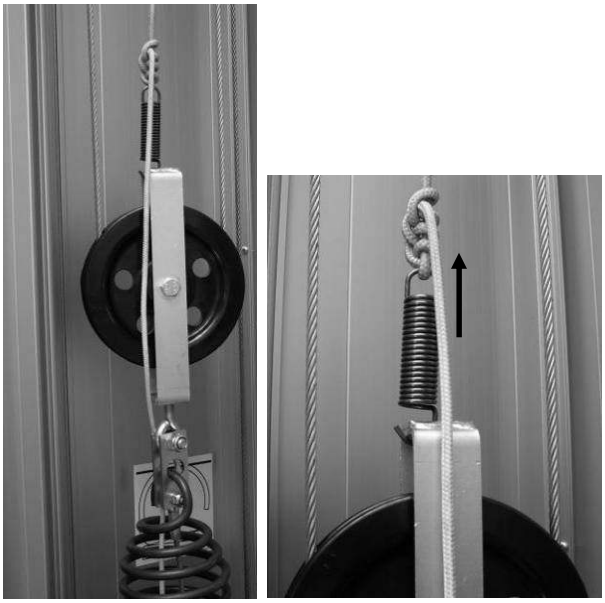


Shows door delivered with BBS

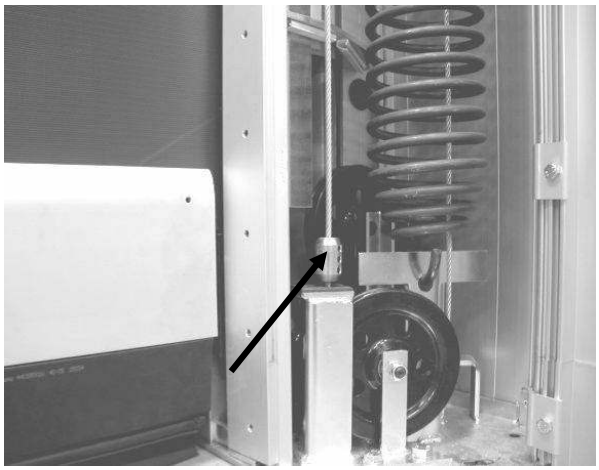


100. The steel rope for BBS and the spring are hooked in (connected) to the pulley (arrow).
Spring system A: Go on with Picture 57
Spring system B: Go on with Picture 73

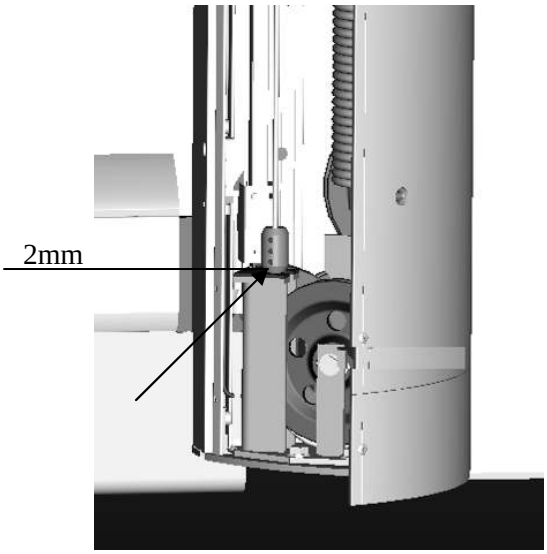
101. Pull the textile cord arround the small pulley at the head plate.



102. Pre-tension the small spring.
Knot the textile cord at the small spring at the pulley (upper spring end).
Close the door with slow mode.



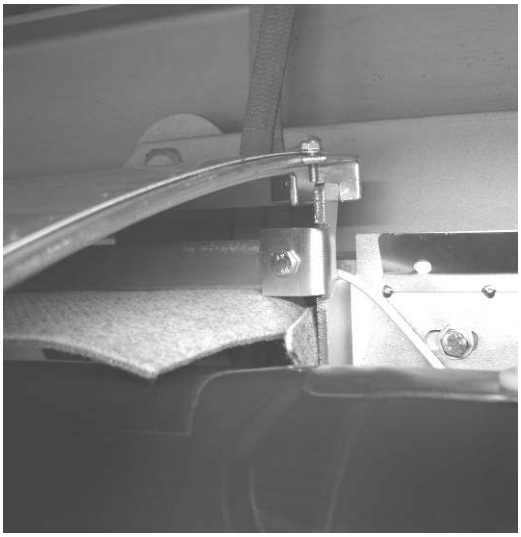
103. Put the cable lock on the steel cable for BBS.



104. Adjust the space between the cable lock and the rubber stopper to maximum 2 mm.

105. Repeat step 101 to 104 for the other side frame.
Maybe it is necessary to adjust the textile cord, if that is the case, make a new knot. Run the door to make sure that the BBS is working properly.

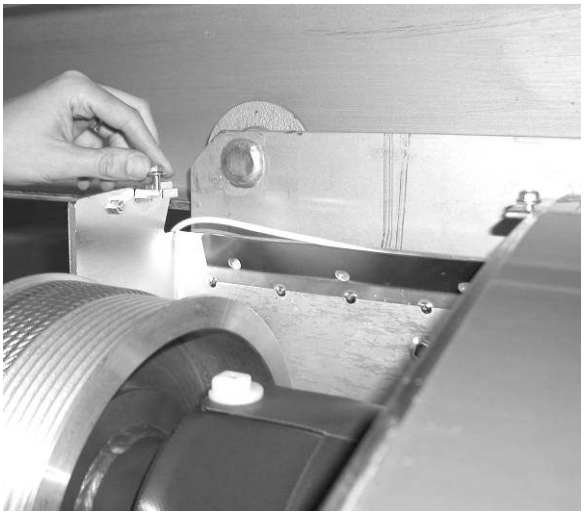
Installation of covers



106. Lift up the top roll cover and hook it on to the upper brackets.



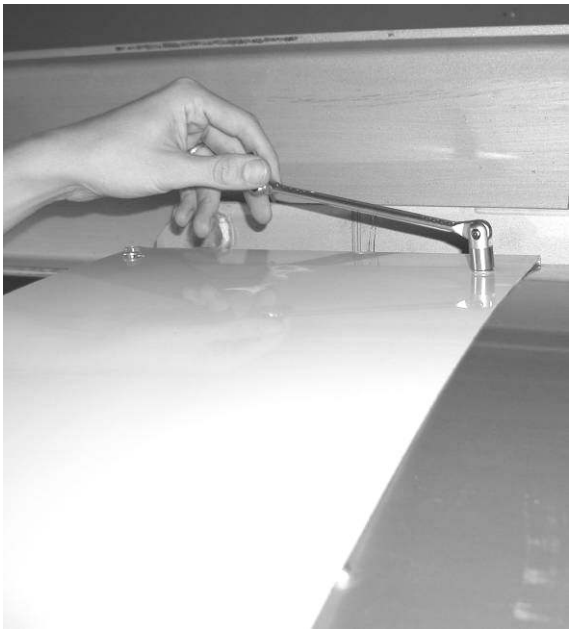
107. Press the lower part in to the slides for the screws and centre. Tighten the screws to lock the top roll cover.



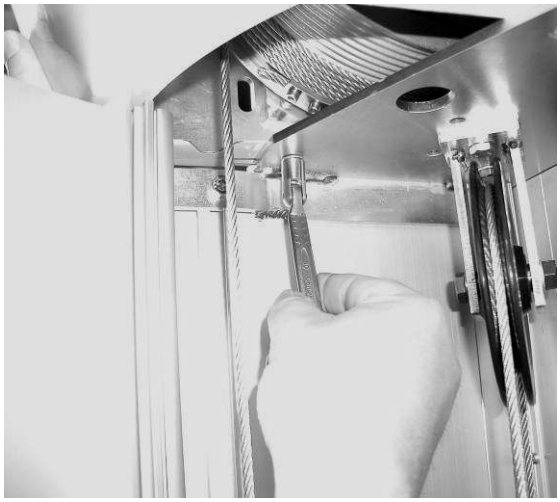
108. Remove the two screws at the top of the wall plate.



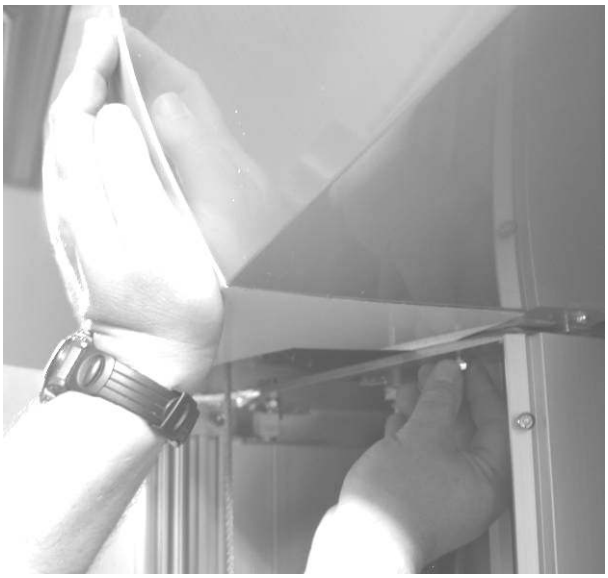
109. Take the cable drum covers and remove the two screws at the bottom.



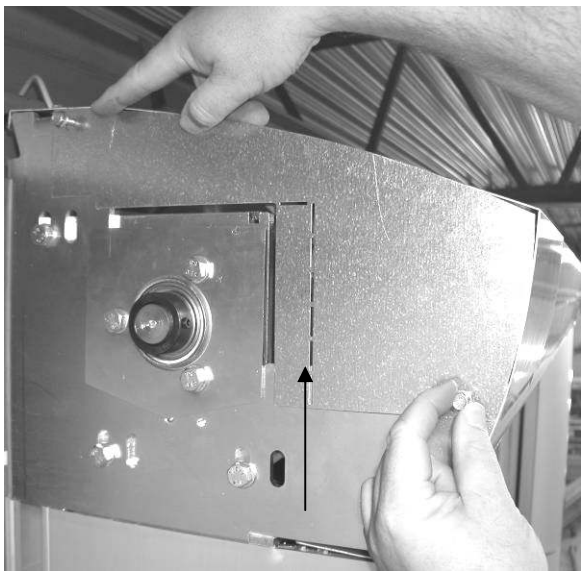
110. Put the cover on the top of the wall bracket and lock it with the two top screws.
These screws also lock the top of the top roll cover.



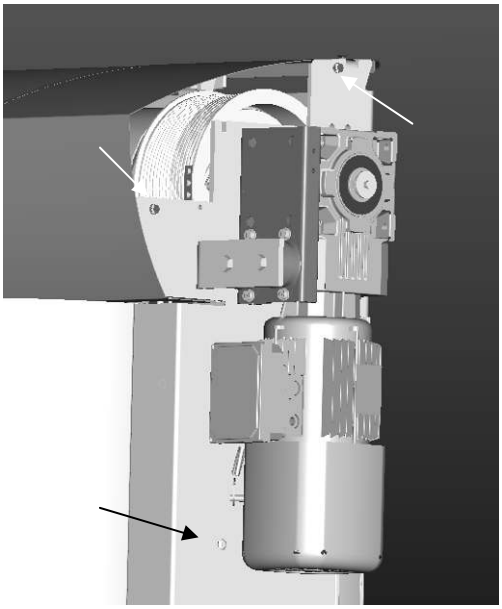
111. Press in the cover at the outside and fit the outer screw.



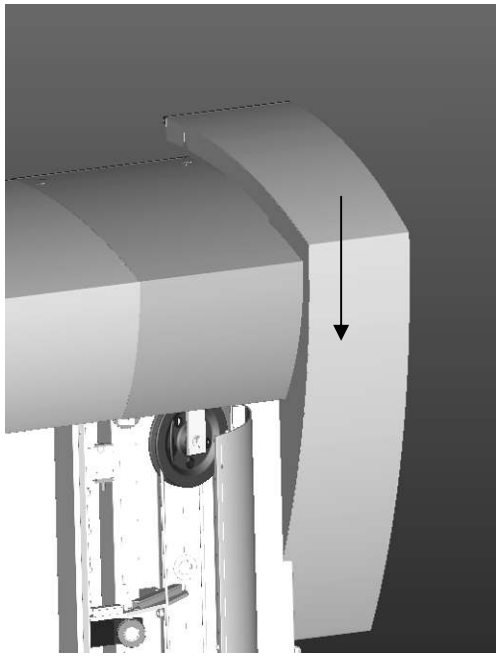
112. Press and form the cover to fit the shape off the top roll cover and lock with the second screw. Do the same on the other side.



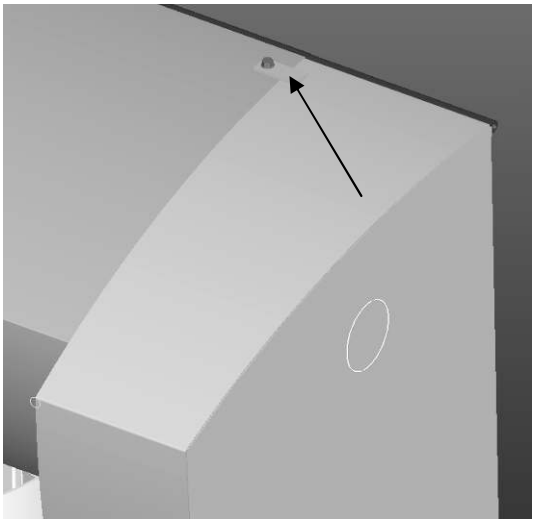
113. Remove the two screws at the non motor side and fit the steel cover as above. If system B is used, cut off the plate extension. If the door has no motor cover but covers for cable drums, the same steel cover must be assembled on the motor side.



114. Lift up the motor cover and fit the slots at the motor cover to the 3 screws marked at the picture…



115. … press down the motor cover.



116. Place the locking bracket over the motor cover.

Dismantling of the door works opposite to the sequence of the installation.

The material of packaging can be given back to the manufacturer.

Spoiled door-components must be wasted by the customer environment-friendly and in harmony with the valid law.

Defect electronic equipment belongs to special waste and because of this specific laws have to be respected in case of disposal.